

EZ-IO® Intraosseous Vascular Access System

Procedure Guidelines and Anesthetic Guideline

Ivinson Memorial EMS

Office of the Medical Director

PURPOSE

To provide procedural guidance for establishment of intraosseous vascular access using the ARROW® EZ-IO® Intraosseous Vascular Access System.

SCOPE OF PRACTICE

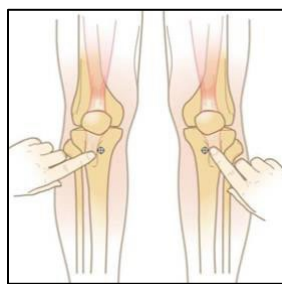
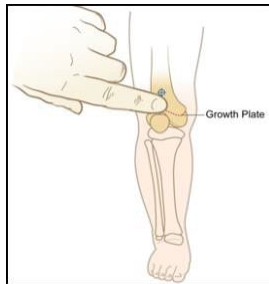
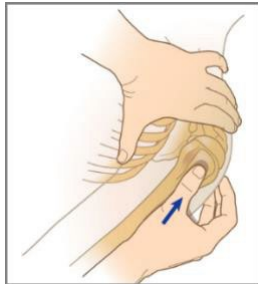
AEMT, IEMT, Paramedic, and RN may insert an IO for Vascular access as needed.

INDICATIONS FOR USE

For adult and pediatric patients any time vascular access is difficult to obtain in emergent, urgent or medically necessary situations. Catheter may remain in patient for up to 24 hours.

Approved IO access sites

Adult & Pediatric
• Proximal humerus • Distal Femur (preferred site if available) • Proximal Tibia • Distal Tibia



CONTRAINDICATIONS

- Fracture of the targeted bone
- Previous, significant orthopedic procedures at insertion site (e.g. prosthetic limb or joint)
- IO in the targeted bone within the past 48 hours
- Infection at area of insertion
- Excessive tissue or absence of adequate anatomical landmarks
 - Consider distal femur cutdown

EQUIPMENT/SUPPLIES

- EZ-IO® Power Driver
- EZ-IO® Needle Set and EZ-Connect® Extension Set
- Gloves
- Cleansing agent

- Luer lock syringe with sterile Normal Saline flush (10 mL for adults, 5 - 10 mL for infant/child)
- Sharps container

ADDITIONAL EQUIPMENT/SUPPLIES IF INDICATED/ORDERED

- 2% Cardiac Intravenous Lidocaine (preservative & epinephrine-free)
- Intravenous fluid
- Infusion pressure pump or pressure bag

PROCEDURE

Explain procedure to patient/family when possible Obtain assistance as needed

INSERTION SITE IDENTIFICATION

Palpate preferred site to locate appropriate anatomical landmarks for needle set placement. Identify soft tissue depth overlying the insertion site.

Utilize the correct technique below based on patient and site selected:

ADULT INSERTION SITE IDENTIFICATION

Distal Femur (Adult)

1. Secure the knee out-stretched to ensure the knee does not bend.
2. Identify the patella by palpation. The insertion site is just proximal to the patella (maximum 2 cm – 2 finger breadths) and approximately 1-2 cm medial to midline. Consider cutdown if excessive tissue

Proximal Humerus (Adult)

1. Adduct humerus
 - Identify the greater tubercle (bump located on the lateral aspect proximal humerus). To feel this prominence on obese patients, palpate deeply.
2. Palpate the surgical neck
 - This may feel like a golf ball on a tee – the spot where the “ball” meets the “tee” is the surgical neck The insertion site is 1 to 2 cm above the surgical neck, on the most prominent aspect of the greater tubercle

Proximal Tibia (Adult)

1. Extend the leg.
2. Insertion site is approximately 2 cm medial to the tibial tuberosity, or approximately 3 cm below the patella and approximately 2 cm medial, along the flat aspect of the tibia.

Distal Tibia (Adult)

1. Insertion site is located approximately 3 cm proximal to the most prominent aspect of the medial malleolus.
2. Palpate the anterior and posterior borders of the tibia to assure insertion site is on the flat center aspect of the bone.

INFANT/CHILD INSERTION SITE IDENTIFICATION

Distal Femur (Infant/Child)

1. Secure the leg out-stretched to ensure the knee does not bend.
2. Identify the patella by palpation. The insertion site is just proximal to the patella (maximum 1 cm – 1 finger breadth) and approximately 1-2 cm medial to midline.

Proximal Humerus (Infant/Child)

1. Place the patient's hand over the abdomen (elbow adducted and humerus internally rotated)

2. Place your palm on the patient's shoulder anteriorly; the "ball" under your palm is the general target area.
 - You should be able to feel this ball, even on obese patients, by pushing deeply
3. Place the ulnar aspect of your hand vertically over the axilla and the ulnar aspect of your other hand along the midline of the upper arm laterally
4. Place your thumbs together over the arm
 - This identifies the vertical line of insertion on the proximal humerus
5. Palpate deeply up the humerus to the surgical neck
 - This may feel like a golf ball on a tee – the spot where the "ball" meets the "tee" is the surgical neck. The insertion site is 1 to 2 cm above the surgical neck, on the most prominent aspect of the greater tubercle

Proximal Tibia (Infant/Child)

1. Extend the leg. Pinch the tibia between your fingers to identify the medial and lateral borders.
2. Insertion site is approximately 1 cm medial to the tibial tuberosity, or just below the patella (approximately 1 cm) and slightly medial (approximately 1 cm), along the flat aspect of the tibia.

Distal Tibia (Infant/Child)

1. Insertion site is located approximately 1-2 cm proximal to the most prominent aspect of the medial malleolus.
2. Palpate the anterior and posterior borders of the tibia to assure insertion site is on the flat center aspect of the bone.

NEEDLE SET SELECTION

Select EZ-IO® Needle Set based on patient weight, anatomy and clinical judgment. The EZ-IO® Catheter is marked with a black line 5 mm proximal to the hub. Prior to drilling, with the EZ-IO® Needle Set inserted through the soft tissue and the needle tip touching bone, adequate needle length is determined by the ability to see the 5 mm black line above the skin.

- EZ-IO® 45 mm Needle Set (yellow hub) should be considered for proximal humerus insertion in patients 40 kg and greater and patients with excessive tissue over any insertion site
- EZ-IO® 25 mm Needle Set (blue hub) should be considered for patients 3 kg and greater
- EZ-IO® 15 mm Needle Set (pink hub) (NOT USED at Iverson Memorial EMS) patients 3-10 kg

INSERTION

Use a clean, "no touch" technique, maintaining asepsis

Prepare supplies

Prepare the site by using antiseptic of your choice; stabilize the extremity

Remove the needle set cap

ADULT INSERTION TECHNIQUE

Femur – Adult

1. Aim the needle set at 90 degree angle to the bone.
2. Push the needle set tip through the skin until the tip rests against the bone

The 5 mm mark must be visible above the skin for confirmation of adequate needle set length
3. Gently drill, advancing the needle set approximately 1-2 cm after entry into the medullary space or until the needle set hub is close to the skin

Proximal Humerus - Adult

1. Aim the needle set at a 45-degree angle to the anterior plane and posteromedial
2. Push the needle set tip through the skin until the tip rests against the bone

The 5 mm mark must be visible above the skin for confirmation of adequate needle set length
3. Gently drill into the humerus approximately 2 cm or until the hub is close to the skin; the hub of the needle set should be perpendicular to the skin

Tibia - Adult

1. Aim the needle set at a 90-degree angle to the bone
2. Push the needle set tip through the skin until the tip rests against the bone

The 5 mm mark must be visible above the skin for confirmation of adequate needle set length

3. Gently drill, advancing the needle set approximately 1-2 cm after entry into the medullary space or until the needle set hub is close to the skin

INFANT/CHILD INSERTION TECHNIQUE

Proximal Humerus – Infant/Child

1. Aim the needle set tip at a 45-degree angle to the anterior plane and posteromedial
2. Push the needle set tip through the skin until the tip rests against the bone

The 5 mm mark must be visible above the skin for confirmation of adequate needle set length

3. Gently drill, immediately release the trigger when you feel the loss of resistance as the needle set enters the medullary space; avoid recoil – do NOT pull back on the driver when releasing the trigger

Femur and Tibia – Infant/Child

1. Aim the needle set at a 90-degree angle to the bone
2. Push the needle set tip through the skin until the tip rests against the bone

The 5 mm mark must be visible above the skin for confirmation of adequate needle set length

3. Gently drill, immediately release the trigger when you feel the loss of resistance as the needle set enters the medullary space; avoid recoil – Do NOT pull back on the driver when releasing the trigger

INSERTION COMPLETION

1. Hold the hub in place and pull the driver straight off; continue to hold the hub while twisting the stylet off the hub with counter clockwise rotations; catheter should feel firmly seated in the bone (1st confirmation of placement);
 - A. Dispose of all sharps and biohazard materials using standard biohazard practices and disposal containers.
 - B. If using the NeedleVISE® 1 port sharps block, place on stable surface and use a one-handed technique.
2. Place the EZ-Stabilizer® Dressing over the hub
3. Attach a primed extension set to the catheter hub, firmly secure by twisting clockwise
4. Pull the tabs off the dressing to expose the adhesive, apply to the skin
5. Aspirate for blood/bone marrow (2nd confirmation of placement)*

**Inability to withdraw/aspirate blood from the catheter hub does not mean the insertion was unsuccessful.*

6. Flush with 10 ml Normal Saline rapid flush

A. ADULT - RESPONSIVE TO PAIN – RECOMMENDED ANESTHETIC

Observe recommended cautions/contraindications to using 2% preservative and epinephrine-free lidocaine (intravenous lidocaine) and confirm lidocaine dose per institutional protocol

Priming volume if the EZ-Connect® Extension Set is approximately 1.0 mL Normal Saline

1. Slowly infuse lidocaine 40 mg IO over 120 seconds
2. Flush Extension set with 2ml saline
3. Allow lidocaine to dwell in IO space at least 60 seconds
4. Flush with 5 to 10 mL of normal saline
5. Slowly administer an additional 20 mg of lidocaine IO over 60 seconds.

Repeat PRN; consider systemic pain control for patients not responding to IO lidocaine

B. INFANT/CHILD - RESPONSIVE TO PAIN – RECOMMENDED ANESTHETIC

Use 2% preservative and epinephrine-free lidocaine (“cardiac” intravenous lidocaine) and confirm lidocaine dose per Handtevy Dosing Guide with initial dose 0.5 mg/kg, not to exceed 40 mg

1. The priming volume of the EZ-Connect® Extension Set is approximately 1.0 mL for such small doses of lidocaine add lidocaine dose to extension set
2. Slowly infuse 2 ml of Normal Saline after the lidocaine to over 120 seconds to infuse the Lidocaine through the extension set and needle into the marrow space.
3. Allow lidocaine with the subsequent saline flush to dwell in IO space 60 seconds

4. Flush with 2-5 mL of normal saline
5. Administer subsequent lidocaine (half the initial dose) IO into extension set followed by 2 ml Normal Saline flush over 60 seconds.

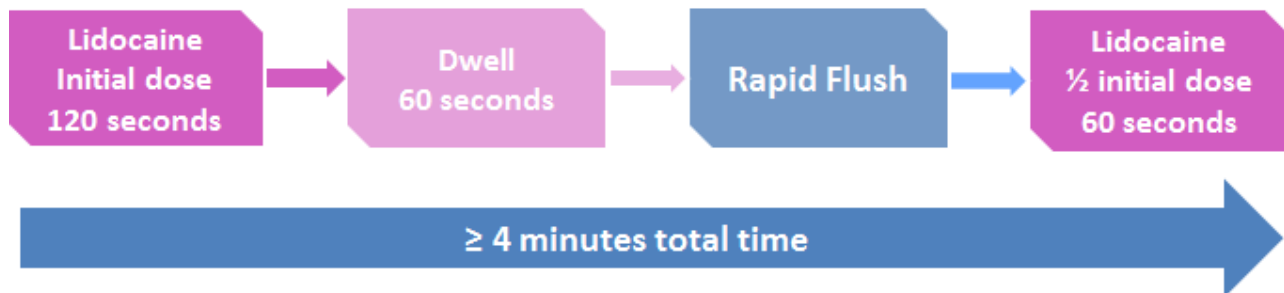
Repeat PRN; consider systemic pain control for patients not responding to IO lidocaine

- Connect fluids if ordered and pressurize up to 300 mmHg for maximum flow
- Verify placement/patency prior to all infusions.
- Stabilize and monitor site and limb for extravasation or other complications
 - For proximal humerus insertions, apply arm immobilizer or other securement device
 - For distal femur insertions, maintain securement of the leg to ensure the **knee does not bend**
- Document date and time on wristband and place on patient

2% preservative- and epinephrine-free lidocaine

Adult: Typically 40mg

Infant/Child: Typically 0.5 mg/kg (NOT to exceed 40 mg)



www.eziocomfort.com

REMOVAL TECHNIQUE

1. Remove extension set and dressing
2. Stabilize catheter hub and attach a Luer lock syringe to the hub
3. Maintaining axial alignment, twist clockwise and pull straight out Do not rock the syringe
4. Dispose of catheter with syringe attached into sharps container
5. Apply pressure to site as needed to control bleeding and apply dressing as indicated

For more information see:

<http://www.teleflex.com/en/usa/ezioeducation/index.html>

Vignette Series - Access one or all of the brief instructional movies in the following live provider/adult patient scenarios:

○ Proximal Humerus Site Identification	View Download
○ EZ IO set UP and Preparation	View Download
○ Proximal Humerus Insertion	View Download
○ Arrow® EZ-Stabilizer® Dressing Application	View Download
○ Recommended Anesthetic Technique	View Download
○ Removal	View Download
○ Proximal Tibia Site Identification & Insertion	View Download